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PII: S0735-1097(26)07589-3

DOI: <https://doi.org/10.1016/j.jacc.2026.08.036>

Reference: JAC 33413

To appear in: JACC

Received Date: 5 August 2026

Revised Date: 24 August 2026

Accepted Date: 25 August 2026

Please cite this article as: Sharma SP, Budde RPJ, Marc van der Zee P, Gürlek C, Muhl C, Dirkali A, Geluk CA, Tanis W, van Gorselen EOF, Oosterhof T, Braam RL, Prado GP-d, van Velzen JE, Yassi W, Wolters SL, Fehér J, Janssen CHC, Jan van Rooden C, Caris RPGJ, Gratama J-WC, Vliegenthart R, van Werkhoven JM, Galema TW, Korthals TS, Mahmoud KD, Smulders MW, Hirsch A, Addition of Computed Tomography-derived Fractional Flow Reserve in the diagnostic pathway of patients with stable coronary artery disease: the FUSION randomized controlled trial, JACC (2026), doi: <https://doi.org/10.1016/j.jacc.2026.08.036>.

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Running title: FFRct in the diagnostic pathway of CAD

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Total word count: 5463

Total tables: 3

Total figures: 5 (including central illustration)

Supplementary tables: 9

Funding

The trial was sponsored by the Dutch Ministry of Health and the Dutch National Health Care Institute (Zorginstituut Nederland, project number 80-86200-98-19012). Additional financial support was received from Heartflow Inc. (San Francisco, California, United States), who had no role in the study design, conduct, data collection, analysis, or reporting.

Disclosures of interest

Institutional support to Erasmus MC by Bracco, Heartflow and Siemens by research grants.

RPJB received speaker fees from Bayer, Bracco, Heartflow and Siemens and served on an advisory board for Bayer, Bracco and Heartflow (payments partially to Erasmus MC).

AH received a research grant and consultancy fees from GE Healthcare and speaker fees from GE Healthcare, Bristol Myers Squibb, Bayer, Siemens Healthineers, and Heartflow. He is a member of the medical advisory board of Medis Medical Imaging Systems.

CM received speakers fee from Bayer. MS and CM are both supported by an Academic Fund grant (MUMC+), and received funding (2024) for a collaborative project co-financed with PPP allowance made available by Health~Holland, Top Sector Life Sciences & Health to stimulate public-private partnerships and funding from Siemens Healthineers Nederland B.V.

RV received speaker fees from Siemens Healthineers, Bayer Healthcare, Keya Medical, and Wiley, and institutional research grants from Siemens Healthineers. KDM has received educational grants from Novartis and Boston Scientific. All other authors report no conflict of interest.

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Acknowledgements

All investigators, research nurses, and members of the clinical event committee deserve recognition for their contributions to the FUSION trial. Additionally, we extend our gratitude to all patients who provided informed consent to participate.

Abstract

Background: Computed Tomography-derived Fractional Flow Reserve (FFRct) integrates anatomical and functional information, which is particularly useful for stable coronary artery disease (CAD) patients with at least intermediate stenosis.

Objectives: We investigated the impact of adding FFRct to the diagnostic pathway of CAD patients with a 50–90% stenosis on Coronary Computed Tomography Angiography (CCTA).

Methods: FUSION is an investigator-initiated, multicenter, randomized controlled trial involving patients with 50–90% stenosis in ≥ 1 coronary artery on CCTA. Patients were randomized to FFRct-guided or usual care. The primary endpoint was invasive coronary angiography (ICA) without obstructive CAD at 90 days. Secondary endpoints included ICA without obstructive CAD, major adverse cardiac events (MACE), and costs at 1 year, as well as revascularizations, and quality of life at 90 days and 1 year. ICA use was assessed as a post-hoc exploratory endpoint.

Results: Overall, 528 patients (median age 63 [57-69] years, 59% male) were randomized to FFRct-guided (n=263) or usual care (n=265). At 90 days, the rate of ICA without obstructive CAD was significantly lower in the FFRct group than in the usual care group (18% (48/263) versus 33% (87/265); odds ratio 0.46; 95% confidence interval, 0.31–0.69; $P < 0.001$). ICA rate was 39% (102/263) versus 51% (136/265), respectively ($P = 0.004$). Both differences persisted at 1 year. Revascularization rates were similar: 20% (52/263) versus 20% (53/265) at 1 year, respectively ($P = 0.948$). Quality of life, costs, and MACE did not differ between groups, although the clinical event rates were low.

Conclusions: Adding FFRct to the diagnostic pathway of patients with a 50-90% stenosis on CCTA reduced the rate of ICA without obstructive CAD at 90 days and 1 year, without differences in revascularizations, quality of life, and costs compared to usual care. Clinical

event rates were similar, although low. The rate of overall ICA use, a post-hoc exploratory outcome, was also reduced at 90 days and 1 year.

Keywords: Computed tomography angiography; Coronary angiography; Coronary artery disease; Fractional flow reserve, Myocardial; Angina, Stable; Costs and cost analysis.

Condensed Abstract

FUSION randomized 528 patients with 50–90% stenosis in ≥ 1 coronary artery on CCTA to FFRct-guided or usual care. At 90 days, ICA without obstructive CAD occurred in 18% versus 33% ($P < 0.001$), and overall ICA in 39% versus 51% (post-hoc exploratory analysis; $P = 0.004$), respectively. These differences persisted at 1 year. Revascularization rates were 20% in both groups. Quality of life, costs, and MACE did not differ between groups, although the clinical event rates were low. Adding FFRct improved diagnostic efficiency by reducing ICA without obstructive CAD and overall ICA use without changing revascularization.

Abbreviations

CAC	Coronary Artery Calcium
CAD	Coronary Artery Disease
CCS	Chronic Coronary Syndrome
CCTA	Coronary Computed Tomography Angiography
CI	Confidence Interval
EQ-5D-5L	EuroQol 5-Dimension 5-Level
FFR	Fractional Flow Reserve
FFRct	Computed Tomography-derived Fractional Flow Reserve
ICA	Invasive Coronary Angiography
MACE	Major Adverse Cardiac Events
OR	Odds Ratio

Introduction

Coronary Computed Tomography Angiography (CCTA) has a class I indication in both American and European guidelines for patients with suspected chronic coronary syndrome (CCS) and a low to moderate pre-test likelihood of obstructive coronary artery disease (CAD).^{1,2} Although CCTA provides high-resolution anatomical assessment of CAD, it lacks the ability to directly evaluate the functional significance of coronary stenoses.³

Consequently, a substantial proportion of patients who are subsequently referred for invasive coronary angiography (ICA) have no hemodynamically significant CAD.³

Computed Tomography-derived Fractional Flow Reserve (FFR_{ct}) can complement CCTA by providing non-invasive, lesion-specific, functional information along the coronary tree.^{4,5}

FFR_{ct} has shown good agreement with invasive Fractional Flow Reserve (FFR).^{6,7} Adding FFR_{ct} to CCTA integrates anatomical and functional information and may reduce the number of ICAs without obstructive CAD. This approach may be particularly useful in patients with >50% anatomical stenosis on CCTA, where a common diagnostic dilemma is whether to proceed directly to ICA, perform additional non-invasive functional testing, or only continue medical therapy.¹

However, prior randomized trials did not directly address the incremental value of FFR_{ct} at this post-CCTA decision point in patients with >50% stenosis in contemporary practice. In the FORECAST and PRECISE trials, randomization preceded CCTA; therefore, the observed effect reflects a diagnostic strategy as a whole rather than the contribution of the FFR_{ct} result itself.^{8,9} The TARGET trial and China CT-FFR Study 3 did randomize after CCTA. However, in TARGET, nearly all patients in the usual care group underwent post-CCTA functional stress testing, predominantly exercise electrocardiography, with ICA generally recommended after a positive result.¹⁰ TARGET therefore evaluated FFR_{ct} against a largely protocolized functional-testing pathway rather than discretionary contemporary management after CCTA.

China CT-FFR Study 3 enrolled a broader CCTA population with 25-99% stenosis and evaluated ICA use as the primary endpoint.¹¹ Thus, this trial did not specifically target the population with the greatest clinical uncertainty regarding referral for ICA in contemporary clinical decision-making (i.e., those with >50% stenosis on CCTA).

In addition, current international guidelines increasingly support CCTA as the first-line test and place more emphasis on optimizing medical therapy first, which may have influenced ICA referral patterns in recent years.^{1,2,12} Whether adding a lesion-specific, off-site, FFRct reduces ICA without obstructive CAD has therefore not been established in the current clinical standard of care in the Western world. Therefore, this pragmatic randomized controlled trial was designed to investigate the impact of adding FFRct to the diagnostic pathway of patients with a 50-90% stenosis on CCTA on the rate of ICA without obstructive CAD.

Methods

Study design

The study was an investigator-initiated, multicenter, randomized, controlled clinical trial in 12 hospitals in the Netherlands enrolling patients with stable CAD and 50–90% stenosis in at least one coronary artery ≥ 2 mm on CCTA. The full rationale and design of the ‘Addition of Computed Tomography-derived Fractional Flow Reserve in the diagnostic pathway of patients with stable coronary artery disease: the FUSION randomized controlled trial’ have been described previously. The trial was registered in the Trial Registration of Medical Research in the Netherlands (NL-OMON54357) on July 27, 2021, before enrollment began on July 28, 2021, and was subsequently registered at ClinicalTrials.gov on December 29, 2021 (NCT05174247).⁴ All patients provided written informed consent. The trial complies

with the Declaration of Helsinki, and the study protocol was approved by the Medical Ethics Committee of Erasmus MC (MEC-2021-0189).

Study participants

Patients (age ≥ 18 years) with a 50-90% stenosis on CCTA in at least one major epicardial vessel with a diameter ≥ 2 mm were approached for inclusion. Patients were excluded if they had unstable angina or acute coronary syndrome, unstable clinical status, expected inability to complete follow-up, a history of coronary revascularization, non-invasive or invasive diagnostic testing for CAD within the last 12 months (except for exercise electrocardiogram), or were unsuitable for revascularization if indicated.

Randomization and procedure

Patients were randomized in a 1:1 ratio to the FFRct (intervention) group or the usual care (control) group using a web-based system (Castor EDC, Amsterdam, The Netherlands). Block randomization was performed using computer-generated, randomly permuted blocks of four, six, or eight, with stratification by study center.

In the FFRct group, the CCTA images were sent to Heartflow Inc. (San Francisco, California, United States) for FFRct analysis. The patient management strategy was determined by incorporating the FFRct results, interpreted in conjunction with the CCTA findings and clinical presentation, including the FFRct value, lesion location, focal versus diffuse disease, and the lesion-specific pressure gradient. If FFRct was negative, deferral from ICA was recommended.

In the usual care group, management was left to the discretion of the treating physician.

Physicians were encouraged to follow the European Society of Cardiology guidelines for the diagnosis and management of CCS.¹³

In both arms, the treating physician made the final decision regarding patient management, without a mandated treatment protocol. Whether or not an invasive functional measurement was performed during ICA was left to the interventional cardiologist.

Measurements and endpoints

The primary endpoint was the rate of ICA without obstructive CAD at 90 days. ICA without obstructive CAD was defined as any ICA without a hemodynamically significant stenosis, defined as an invasive FFR ≤ 0.80 or an instantaneous wave-free ratio ≤ 0.89 . In the absence of invasive functional measurements, ICA without obstructive CAD was assessed using quantitative coronary angiography (no percentage diameter stenosis $\geq 50\%$) when deemed necessary by the clinical events committee, or by visual estimation (no percentage diameter stenosis $\geq 50\%$ for left main and $\geq 70\%$ stenosis for all other segments) when quantitative coronary angiography was not feasible.⁴ All original ICA images were scored by the blinded clinical event committee.

The primary analysis was performed according to the intention-to-treat principle. In addition, a per-protocol analysis was performed, excluding patients in the FFRct group for whom FFRct results were not incorporated into clinical decision-making. These were defined as patients for whom the treating physician had already determined the treatment plan before the FFRct result was available or who failed to take the available FFRct result into account. A third analysis was performed, also excluding patients in whom the FFRct was not feasible. Secondary outcomes were all exploratory and included: (1) the rate of ICA without obstructive CAD (1 year); (2) coronary revascularizations (90 days/1 year); (3) additional non-invasive diagnostic tests for CAD assessment (90 days/1 year); (4) major adverse cardiac events (MACE), defined as a composite of all-cause mortality, non-fatal myocardial infarction, and unplanned hospitalization leading to urgent revascularization (1 year); (5)

cardiovascular mortality (1 year); (6) non-fatal stroke (1 year); (7) complications during or after ICA (1 year); (8) quality of life and angina status (baseline/90 days/1 year); and (9) CAD-related healthcare costs (1 year) as described in the following paragraph. Overall ICA use at 90 days and 1 year was assessed as a post-hoc exploratory outcome. Data were collected using direct patient contact by research staff at each center, as well as from local healthcare records. All ICAs and clinical events were reviewed and adjudicated by a blinded clinical events committee. Quality of life and angina status were assessed using the EuroQol 5-Dimension 5-Level (EQ-5D-5L), Short Form Health Survey 36, and the Seattle Angina Questionnaire.⁴ A retrospective blinded FFRct analysis in the usual-care group was prespecified. These analyses are ongoing, not included in the present report, and will be reported separately.

Cost analysis at 1 year

CAD-related healthcare costs per patient at 1-year follow-up were calculated by multiplying healthcare use between randomization and 1-year follow-up by resource-specific cost prices. The medical resource use in this calculation comprised diagnostic testing for CAD (i.e. stress echocardiogram or electrocardiogram, positron emission tomography myocardial perfusion imaging, single photon emission computed tomography, cardiovascular magnetic resonance perfusion imaging or CCTA), ICA, percutaneous coronary intervention, coronary artery bypass grafting, outpatient visits to a cardiologist, CAD-related hospital admission, and cardiovascular medication. Data on healthcare resource use were collected during scheduled follow-up visits and from electronic patient records. Data on the number of outpatient visits were not routinely collected. Therefore, based on expert opinion, it was assumed that the intervention would not affect the average number of visits per patient and that patients in both study groups would, on average, have three visits during the 1-year follow-up. If available,

cost prices were based on the Costing Manual of the Dutch National Healthcare Institute and the cost price benchmark of the University Medical Centers Netherlands. Several cost prices were merged, using weighted and aggregated cost estimates from multiple sources. An overview of medical resource use and corresponding cost prices is provided in Supplementary Table S1. All costs were indexed to 2025 Euros using the consumer price index from the Dutch Central Bureau of Statistics and were not discounted because of the short follow-up period. At the beginning of the study, the cost price of an FFRct analysis was set at €1,000. A sensitivity analysis was performed with a cost price of €600.

Sample size

Based on previous studies, 35% of patients referred for ICA after CCTA were expected to have no hemodynamically significant stenosis.^{14,15} The addition of FFRct was expected to reduce the number of ICAs by 33%, corresponding to an expected rate of ICA without obstructive CAD of 23% in the FFRct group versus 35% in the usual care group. To reach a power of 80% with a two-sided significance level of 0.05, 1:1 randomization, and assuming a 10% dropout rate, 528 patients were required.⁴

Statistical analysis

Continuous data are presented as medians (25th-75th percentiles) and categorical data as frequencies.

ICA without obstructive CAD, ICA use, revascularizations, and complications after ICAs were analyzed using logistic regression, with randomization group as the independent variable. Results are reported as odds ratios (ORs) with 95% confidence intervals (CIs). Prespecified multivariable logistic regression analyses, adjusted for age, sex, type of chest pain, and coronary artery calcium (CAC) score, were performed to assess ICA without

obstructive CAD at 90 days and 1 year. Sex and type of chest pain were entered as categorical variables, with male sex and typical chest pain as the reference categories. Given the skewed distribution of CAC scores, CAC was entered as a natural log-transformed variable [$\ln(\text{CAC} + 1)$]. One missing CAC value, due to unavailable non-contrast CT acquisition, was imputed using the median CAC score of the study cohort. Finally, mixed-effects logistic regression analyses were performed, including the same variables as fixed-effects parameters as the previous model, however with a random intercept for center.

Cumulative MACE rates were estimated using the Kaplan-Meier method. A hazard ratio with 95% CI was estimated using a Cox proportional hazards model, with randomization group as the only covariate. No multivariable analysis was performed due to the low number of events. EQ-5D-5L index values were calculated using the Dutch tariff.¹⁶ The Seattle Angina Questionnaire summary score was calculated as the mean of the physical limitation, angina frequency, and quality-of-life domains. Short Form Health Survey 36 domain scores were transformed to a 0–100 scale according to the RAND-36 scoring method, with higher scores indicating better health status.¹⁷ Change from baseline was calculated for each questionnaire outcome as the follow-up score minus the baseline score. Questionnaire scores at each timepoint and changes from baseline were compared between the FFRct and usual care groups using Mann–Whitney U tests. No formal imputation methods were used for missing questionnaire data.

The mean CAD-related healthcare costs per patient are presented with 95% CIs that were estimated using a nonparametric bootstrap approach with 10,000 repetitions.

All P-values are two-sided, and statistical significance was set at 0.05. Statistical analyses were performed using IBM SPSS Statistics for Windows version 28 (IBM Corp., Armonk, NY) and R version 4.5.2 (R Foundation for Statistical Computing, Vienna, Austria).

Results

Between July 2021 and July 2025, 528 patients were randomly assigned to the FFRct group (263) or the usual care group (265)(Figure 1). Among included patients, 59% were male, median age was 63 [57-69] years, and 62% had typical or atypical chest pain. Baseline characteristics were well balanced between the two arms (Table 1). At 1-year follow-up, no patients were lost to follow-up.

CCTA

The extent and severity of CAD assessed by CCTA, based on local site interpretation, were comparable between the two arms. Two- or three-vessel disease and/or left main disease with a diameter stenosis $\geq 50\%$, was present in 32% (85/263) of patients in the FFRct group and 30% (80/265) in the usual care group (Table 1). In 43% (114/263) of the patients in the FFRct group and 46% (122/265) in the usual care group, the left main and/or proximal left anterior descending artery was involved.

Diagnostic strategy after CCTA

In the FFRct arm, FFRct analysis was successfully performed in 91% (240/263) of patients. In the remaining 9% (23/263), FFRct processing was not feasible due to image quality or technical constraints, including motion (n=9), misalignment (n=7), noise (n=4), low contrast (n=1), oversized field of view (n=1), or other technical issues (n=1). In 3% (7/240) of the patients with available FFRct results, the findings were not incorporated into the clinical decision-making process.

Primary endpoint

At 90 days, the rate of ICA without obstructive CAD was significantly lower in the FFRct group than in the usual care group (18% [48/263] vs 33% [87/265]; OR 0.46; 95% CI, 0.31–0.69; $P<0.001$)(Table 2; Figure 2). In the adjusted models, the effect size remained consistent (Table 2). In the per-protocol analysis, the unadjusted OR was 0.39 (95% CI, 0.26–0.59; $P<0.001$), and in the sensitivity analysis, excluding also patients with missing FFRct results in addition to the per-protocol analysis, it was 0.35 (95% CI, 0.22–0.55; $P<0.001$). Results from the adjusted analyses are presented in Supplementary Table S2.

Secondary endpoints

At 1-year follow-up, the rate of ICA without obstructive CAD remained significantly lower in the FFRct group compared to the usual care group (22% [57/263] vs 39% [103/265]; OR 0.44; 95% CI, 0.30–0.64; $P<0.001$)(Table 2). Overall ICA use (a post-hoc exploratory endpoint) was also significantly lower in the FFRct group, both at 90 days (39% [102/263] vs 51% [136/265]; OR 0.60; 95% CI, 0.43–0.85; $P=0.004$) and at 1 year (43% [114/263] vs 61% [161/265]; OR 0.49; 95% CI, 0.35–0.70; $P<0.001$)(Table 2; Figure 2). Exploratory subgroup analyses by CCTA characteristics are presented in Supplementary Table S3 and are purely descriptive.

Among patients who underwent ICA at 90 days, obstructive CAD was present in 53% [54/102] in the FFRct group and in 36% [49/136] in the usual care group ($P=0.01$).

Cumulatively at 1 year, obstructive CAD was present in 50% [57/114] and 36% [58/161], respectively ($P=0.02$).

Whether ICA did not show obstructive CAD was adjudicated using invasive FFR or instantaneous wave-free ratio in 41% (112/275) of ICA assessments and by quantitative coronary angiography or visual assessment in 59% (163/275). Invasive FFR or instantaneous

wave-free ratio was used in 44% (50/114) of ICA assessments in the FFRct group and 39% (62/161) in the usual care group.

The proportion of patients undergoing coronary revascularization within 1 year was 20% [52/263] in the FFRct group and 20% [53/265] in the usual care group ($P=0.948$); percutaneous coronary intervention was performed in 45 and 47 patients, respectively, and coronary artery bypass grafting in 7 and 6 patients, respectively. Likewise, the proportion of patients undergoing at least one additional non-invasive diagnostic test did not differ significantly between groups (11% [28/263] vs 14% [37/265]; $P=0.246$)(Supplementary Table S4).

Clinical event rates were low and comparable between groups (Table 3; Figure 3). MACE occurred in 3% [7/263] of patients in the FFRct group and 2% [6/265] in the usual care group ($P=0.766$).

Figure 4 shows subsequent management and outcomes in the usual care group and FFRct group stratified by FFRct results (positive/negative/not feasible). Notably, 35% [17/48] of ICAs without obstructive CAD at 90 days in the FFRct group occurred in patients with a negative or missing FFRct. Furthermore, among patients with a negative FFR ($n=96$), 1 underwent PCI at 1 year, and no MACE occurred.

Supplementary Table S5 presents medication use at baseline, 90 days, and 1 year. Use of lipid-lowering, antiplatelet, and anticoagulation medications was comparable between randomization groups.

Quality of life and angina status

Patient questionnaire completion rates varied across the three timepoints and by the type of survey and ranged from 73% to 92% in the FFRct group and 72% to 87% in the usual care group. EQ visual analog scale, EQ-5D-5L index values, Seattle Angina Questionnaire

summary scores, and the Short Form Health Survey 36 scales were comparable between the FFRct and usual care groups at baseline, 90 days, and 1 year. Overall, changes in quality of life and angina status over the follow-up period did not differ significantly between the FFRct group and the usual care group (Supplementary Tables S6 and S7).

Cost analysis at 1 year

The mean CAD-related healthcare costs per patient in the FFRct group were €3,842 (95% CI, €3,408 - €4,391) versus €3,415 (95% CI, €3,006 - €3,984) in the usual care group. The estimated mean difference in costs between the two groups was €427 (95% CI, €-272 - €1,077). An overview of costs by resource category is provided in Supplementary Table S8. In the sensitivity analysis with a FFRct price of €600, the estimated mean difference was €62 (€-639 - €717)(Supplementary Table S9).

Discussion

In the FUSION trial, adding FFRct to the diagnostic pathway for patients with 50-90% stenosis on CCTA significantly reduced the rate of ICA without obstructive CAD at both 90 days and 1 year, with similar revascularization rates. No differences were observed in quality of life, costs, and clinical events, although the clinical event rates were low. The rate of overall ICA use, a post-hoc exploratory outcome, was also reduced at 90 days and 1 year. Patients with a 50-90% stenosis on CCTA represent a clinically challenging population, as anatomical stenosis severity alone does not reliably identify flow-limiting lesions and often warrants further functional assessment.¹⁸ FUSION addresses whether FFRct-guided care offers incremental value in patient selection for ICA in an era in which guideline-directed first-line testing has increasingly shifted towards CCTA, alongside greater emphasis on initial medical therapy in the management of patients with CAD.

Previous studies have also suggested that FFRct-guided strategies may improve patient selection but evaluated FFRct as part of a broader diagnostic pathway rather than isolating its incremental value after CCTA. PLATFORM was not randomized and enrolled patients already scheduled for ICA.¹⁹ Consequently, the reduction in ICA cannot be causally attributed to FFRct and does not directly address the clinical decision of whether to refer a patient for ICA after an intermediate stenosis has been identified on CCTA. FORECAST and PRECISE randomized patients before CCTA and therefore likewise could not isolate the effect of FFRct.⁸ In a predefined sub-analysis of FORECAST restricted to patients for whom the physician had already chosen CCTA as the first-line test, the reduction in ICA was no longer apparent (17% versus 19%, $P=0.55$), and the rate of ICA without obstructive CAD was not reported.²⁰ In PRECISE, FFRct was part of a diagnostic strategy for patients with stable symptoms and no known CAD. The strategy used the PROMISE minimal-risk score to defer testing in low-risk patients and CCTA with selective FFRct (for 30-90% stenosis) in all others.⁹ These trials therefore primarily provide insight into how the overall diagnostic pathway for patients with chest pain may be organized, whereas FUSION evaluates whether FFRct improves diagnostic efficiency after >50% stenosis has been identified on CCTA. In the TARGET trial and the China CT-FFR Study 3, patients were, however, randomized after CCTA.^{10,11} In TARGET, usual care consisted of protocolized stress testing, predominantly exercise electrocardiography, with ICA recommended after a positive test. The trial therefore compared two functional tests after CCTA rather than FFRct with the contemporary standard of care. TARGET was conducted in six tertiary Chinese centers and showed that on-site FFRct reduced ICA without obstructive CAD, although ICA use was relatively high in both groups (69.2% and 79.4%). China CT-FFR Study 3 enrolled patients with 25-99% stenosis, 56% of whom had no obstructive CAD on CCTA, and therefore did not specifically target patients in whom FFRct is intended to guide the post-CCTA decision to

proceed to ICA. Consistent with this lower-risk population, ICA use at 90 days was low (10.0% versus 12.4%).

In both trials, the use of invasive physiology during ICA to determine obstructive CAD was rare (11% in TARGET and 1.2% in China CT-FFR Study 3), whereas invasive physiology was used in 41% of ICAs in FUSION. These differences likely reflect differences in local practice and trial design. Revascularization patterns also differed between the trials. In TARGET, revascularization increased with FFRct (49.7% versus 42.8% at 90 days), in a setting where FFRct replaced predominantly exercise electrocardiography, and most patients ultimately underwent ICA. In the China CT-FFR Study 3, the revascularization rate was lower in the FFRct arm (7.7% versus 9.5% at 1 year), largely because fewer patients underwent ICA. Among patients who underwent ICA, the proportion of revascularized patients was similar (67.3% versus 67.6%), and the proportion of ICAs without obstructive CAD did not differ.

The reduction in ICAs in FUSION was selective: fewer ICAs were performed in the FFRct arm, while the percentage of patients with obstructive CAD on ICA in the total cohort was comparable (22% vs 22%), increasing the diagnostic yield of ICA from 36% to 50%, while revascularization was unchanged (20% versus 20%). This effect was achieved using an off-site FFRct solution in academic and referral hospitals.

Consistent with previous FFRct trials, the 1-year MACE rate did not differ between the FFRct-guided care and usual care. However, the clinical event rates were low, and FUSION was not powered to detect differences in MACE.

Although we hypothesized that FFRct-guided care would reduce downstream resource use and costs, neither the rate of additional non-invasive testing nor total costs differed significantly between the two strategies. Implementing FFRct did not yield cost savings, with a numerically higher estimated mean cost in the FFRct group of €427 (95% CI, €-272 -

€1,077), despite fewer ICAs. This is consistent with the FORECAST trial, in which costs were the primary endpoint and a diagnostic strategy of CCTA with selective FFRct also did not reduce healthcare costs. In FUSION, cost savings from fewer ICAs did not offset the costs of FFRct, as the price difference in our analysis between an ICA (€1,255) and an FFRct analysis (€1,000) is relatively small. Our sensitivity analysis showed that lowering the FFRct costs to €600 resulted in a lower estimated mean cost difference of €62 (95% CI, €-639 - €717) but still no reduction in costs. Most patients in the usual care arm were referred directly to ICA without an additional non-invasive stress test. Another diagnostic strategy including other non-invasive functional testing after CCTA may be more cost-effective.

Several other aspects of the cost analysis are important. First, the cost calculation included only the prices of university medical centers in the Netherlands. These cost prices may differ from those in non-academic hospitals. Second, the economic findings should be interpreted in the context of the Dutch healthcare system, as costs, reimbursement structures, and the price of FFRct analysis may differ across countries. Third, outpatient visits were not routinely collected and were therefore modeled as three cardiology outpatient visits per patient in both groups, based on expert opinion. This assumption may have influenced the cost analysis, as FFRct-guided care could alter the frequency of downstream clinical consultations. Finally, costs were calculated from an in-hospital health care perspective, overlooking relevant costs such as informal care costs, extramural costs, and productivity loss. However, beyond cost, from a patient perspective, reducing ICAs without obstructive CAD remains relevant, as it reduces exposure to an invasive procedure and its associated procedural burden and risks.

Quality of life and angina-related health status did not differ during follow-up in either group, with no consistent differences between the FFRct-guided care and usual care. This finding aligns with previous studies, including the FORECAST and PLATFORM trials.^{8,21} One possible explanation is that physicians actively treated anginal symptoms in both groups,

primarily with optimal medical therapy, reserving revascularization for selected patients. This approach is supported by the ISCHEMIA trial, which underscored the central role of medical therapy in the management of stable CAD.²²

To translate the findings of our study into routine practice, adequate clinical interpretation of FFRct in local hospitals is essential. In FUSION, the radiologists and cardiologists of the 12 participating hospitals were trained to interpret the FFRct report in the context of the CCTA findings, patient symptoms, lesion location and length, and the presence of focal or diffuse CAD and FFRct delta. To maximize patient benefit, the final treatment decision should be based on this integrated assessment rather than on the FFRct value alone. Wider adoption of FFRct further depends on local implementation and reimbursement structures.

In the FUSION trial, we interpreted FFRct results alongside CCTA findings. However, the acquired CCTA segmentation may provide additional parameters, such as plaque volume and characterization, whose additional value in patient care still needs to be established.^{23,24}

Future studies should also clarify at which time point within a CCTA-first pathway FFRct analyses should be performed: routinely in all patients with intermediate stenosis or selectively for patients with persistent symptoms despite optimal medical therapy in whom revascularization is being considered. Finally, the prognostic value of serial changes in lesion-specific FFRct warrants further investigation, particularly in patients in whom invasive testing is deferred but who may remain at risk for progression of non-significant lesions.

The FUSION study has some limitations. First, not all ICAs were assessed using invasive FFR or instantaneous wave-free ratio. This was inherent to the pragmatic design and reflects routine clinical practice. Because anatomical stenosis severity does not always correspond to hemodynamic significance, this may have resulted in misclassification of the primary endpoint. However, the proportion of ICAs assessed with invasive physiology was comparable between groups (44% vs 39%), reducing the likelihood of substantial bias.

Second, optimization of medical therapy was not standardized in the study protocol. Third, the final treatment plan was left to the treating physician's discretion. Consequently, in 3% of patients with available FFRct results at the time of treatment planning, the findings were not incorporated into clinical decision-making. Nevertheless, the treatment effect remained robust in the intention-to-treat analysis, and a per-protocol analysis was performed. Fourth, the study was conducted in the Netherlands in a predominantly Caucasian population, which should be considered when interpreting the generalizability of the findings. Finally, the study was not designed to determine differences in MACE, and clinical event rates were low.

In conclusion, adding FFRct to the diagnostic pathway of patients with 50-90% stenosis on CCTA reduced the rate of ICA without obstructive CAD at 90 days and 1 year, without differences in revascularizations, quality of life, and costs compared to usual care. Clinical event rates were similar, although low. The rate of overall ICA use, a post-hoc exploratory outcome, was also reduced at 90 days and 1 year.

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Figure legends

Figure 1. Study flowchart.

CAD: coronary artery disease; CCTA: Coronary Computed Tomography Angiography;
FFRct: Computed Tomography-derived Fractional Flow Reserve.

Figure 2. Unnecessary invasive coronary angiography, any invasive coronary angiography, and coronary revascularization within 90 days and 1 year.

P-values are from unadjusted logistic regression. Revascularization included percutaneous coronary intervention or coronary artery bypass grafting.

FFRct: Computed Tomography-derived Fractional Flow Reserve; ICA: invasive coronary angiography.

Figure 3. Kaplan-Meier survival curve of the cumulative incidence of major adverse cardiac events.

Major adverse cardiac events (MACE) included all-cause mortality, non-fatal myocardial infarction, and unplanned hospitalization leading to urgent revascularization.

CI: confidence interval; FFRct: Computed Tomography-derived Fractional Flow Reserve;
HR: hazard ratio.

Figure 4. Downstream clinical management and outcomes stratified by randomization group and FFRct results.

Patients randomized to the FFRct group are stratified by FFRct results: positive FFRct (≤ 0.80), negative FFRct (> 0.80), or FFRct not feasible because of technical or image-quality constraints.

CABG: coronary artery bypass grafting; CAD: coronary artery disease; ICA: invasive coronary angiography; iFR: instantaneous wave-free ratio; FFR: Fractional Flow Reserve; FFRct: Computed Tomography-derived Fractional Flow Reserve; MACE: major adverse cardiac event; MI: myocardial infarction; PCI: percutaneous coronary intervention; Revasc: revascularization.

Central illustration. FFRct in the diagnostic pathway of coronary artery disease: the FUSION Trial.

CAD: coronary artery disease; CCTA: Coronary Computed Tomography Angiography; CI: confidence interval; FFRct: Computed Tomography-derived Fractional Flow Reserve; ICA: invasive coronary angiography; OR: odds ratio.

Perspectives

Competency in Patient Care: In stable CAD patients with 50–90% stenosis in ≥ 1 coronary artery on CCTA, FFRct-guided care improved diagnostic efficiency by reducing the rate of ICA without obstructive CAD, without differences in revascularizations. The rate of overall ICA use, a post-hoc exploratory outcome, was also reduced.

Translational Outlook: Further studies should define the impact of FFRct-guided care on clinical events and determine the optimal timing of FFRct within a CCTA-first pathway, including whether FFRct should be performed routinely after CCTA or selectively after first optimizing medical therapy.

Tables

Table 1. Baseline characteristics

Baseline characteristics	FFRct (n=263)	Usual care (n=265)
Age (years)	63 [57-70]	63 [58-68]
BMI (kg/m ²)	27 [24-29]	26 [24-29]
Gender (male)	160 (61)	152 (57)
Ethnicity		
Caucasian	239 (91)	237 (89)
Asian	3 (1)	4 (2)
South-Asian Surinamese	9 (3)	12 (5)
African Surinamese	3 (1)	4 (2)
Turkish	7 (3)	6 (2)
Other	2 (1)	2 (1)
Smoking		
Never	124 (47)	139 (52)
Former	101 (38)	85 (32)
Current	38 (14)	41 (16)
Diabetes mellitus	40 (15)	39 (15)
Hypertension	129 (49)	132 (50)
Hypercholesterolemia	125 (48)	144 (54)
Familial hypercholesterolemia	7 (3)	3 (1)
Family history of CAD	100 (38)	104 (39)
Previous myocardial infarction	1 (0.4)	3 (1)
History of atrial fibrillation/flutter	32 (12)	35 (13)
Primary presenting symptoms		
Typical	66 (25)	75 (28)
Atypical	104 (40)	84 (32)
Non-anginal	51 (19)	58 (22)
Angina equivalent	32 (12)	35 (13)
Unknown	10 (4)	13 (5)
Medication prior to first outpatient visit		
Lipid-lowering	129 (49)	131 (49)
Antiplatelet*	71 (27)	83 (31)
Anticoagulation**	36 (14)	34 (13)
Coronary Computed Tomography Angiography		
Coronary artery calcium score	229 [103-521]	227 [60-464]
Obstructive CAD based on $\geq 50\%$ stenosis		
One vessel	178 (68)	185 (70)
Two vessels	59 (22)	51 (19)
Three vessels and/or left main	26 (10)	29 (11)
Left main and/or proximal left anterior descending artery	114 (43)	122 (46)
FFRct results		
Hemodynamically significant focal lesion on FFRct (≤ 0.80)***	144 (60)	NA
Data are presented as median [25 th – 75 th percentile] or frequency (percentage).		
*Including salicylates and P2Y12 receptor inhibitors. ** Including direct oral anticoagulants and vitamin K antagonist. *** Out of 240 available analyses.		
CAD: coronary artery disease; FFRct: CT-derived Fractional Flow Reserve. NA: not applicable.		

Table 2. Primary endpoint, diagnostic and procedural outcomes

Endpoint or clinical event	FFRct (n=263)	Usual care (n=265)	Unadjusted OR (95% CI)	P-value	Adjusted OR (95% CI)*	P-Value	Adjusted OR (95% CI)**	P-Value
Primary endpoint								
ICA without obstructive CAD at 90 days	48 (18)	87 (33)	0.46 (0.31-0.69)	<0.001	0.47 (0.31-0.71)	<0.001	0.44 (0.29-0.67)	<0.001
Secondary endpoints								
ICA without obstructive CAD at 1 year	57 (22)	103 (39)	0.44 (0.30-0.64)	<0.001	0.45 (0.30-0.66)	<0.001	0.42 (0.28-0.63)	<0.001
ICA at 90 days [§]	102 (39)	136 (51)	0.60 (0.43-0.85)	0.004	-		-	
ICA at 1 year [§]	114 (43)	161 (61)	0.49 (0.35-0.70)	<0.001	-		-	
Revascularization at 90 days	42 (16)	41 (15)	1.04 (0.65-1.66)	0.875	-		-	
Revascularization at 1 year	52 (20)	53 (20)	0.99 (0.64-1.51)	0.948	-		-	
Complications during or after ICA at 1 year [#]	3 (1)	3 (1)	1.01 (0.20-5.04)	0.993	-		-	

Data are presented as frequency (percentage) or as odds ratio with 95% confidence interval.

*Odds ratio was adjusted for age, sex, ln(coronary artery calcium score + 1), and type of chest pain.

**Odds ratio was adjusted for age, sex, ln(coronary artery calcium score + 1), type of chest pain, and center.

[§] Post-hoc exploratory endpoint.

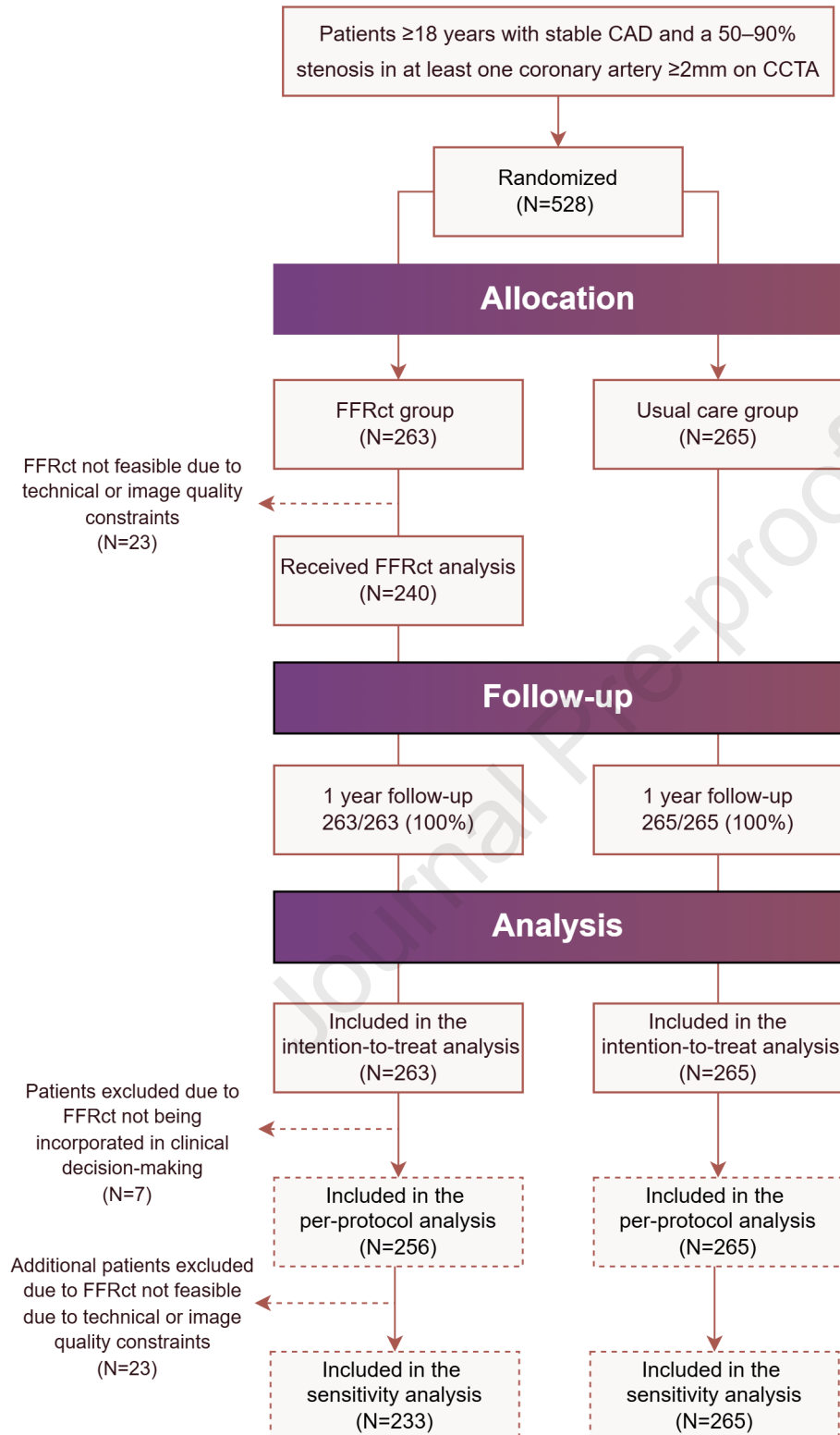
[#] Complications during or after ICA include any complications that occurred within the cardiac catheterization laboratory or ≤ 72 hours after the index procedure, defined as procedural death, stroke, or emergent cardiac surgery, coronary complications, contrast reactions, acute kidney injury, or acute access-site injury.

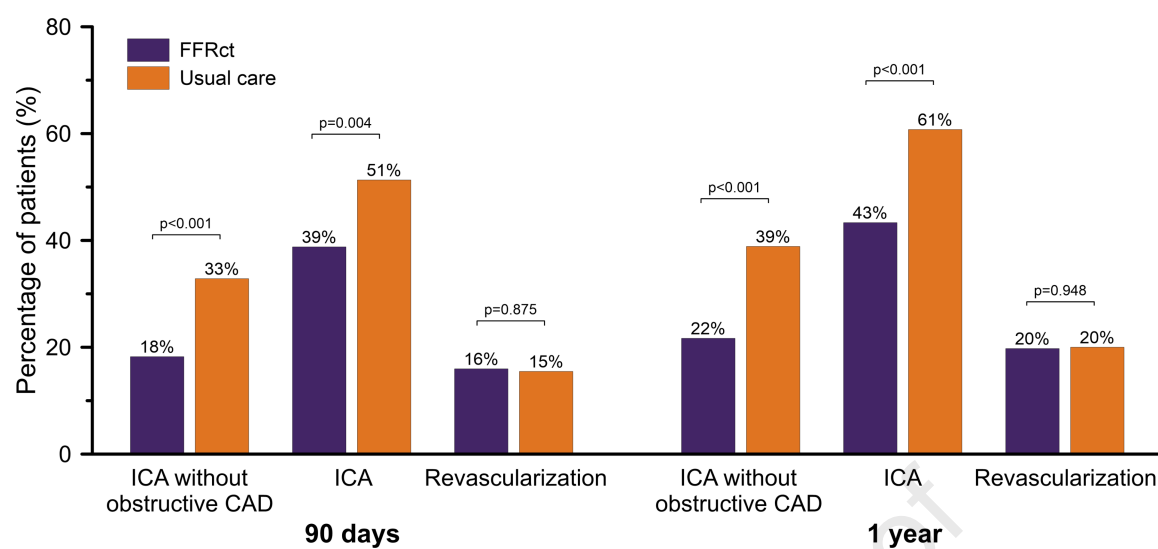
CI: confidence interval; FFRct: Computed Tomography-derived Fractional Flow Reserve; ICA: invasive coronary angiography; OR: odds ratio.

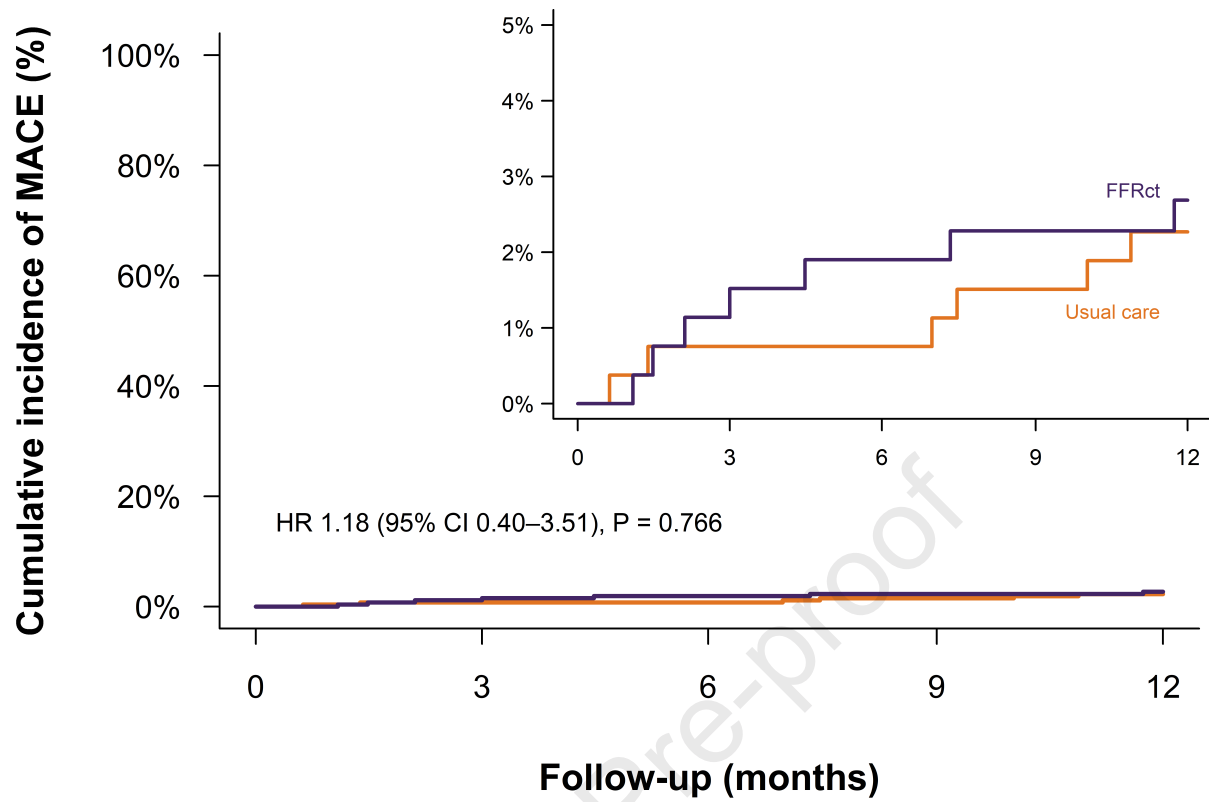
Table 3. Clinical outcomes at 1 year

Endpoint or clinical event	FFRct (n=263)	Usual care (n=265)	Hazard ratio (95% CI)	P-value
At least one MACE	7 (3)	6 (2)	1.18 (0.40-3.51)	0.766
All-cause mortality	4 (2)	1 (0.4)	-	-
Non-fatal myocardial infarction	3 (1)	5 (2)	-	-
Unplanned hospitalization leading to urgent revascularization	2 (1)	4 (2)	-	-
Cardiovascular mortality	1 (0.4)	0 (0)	-	-
Non-fatal stroke	2 (1)	0 (0)	-	-
Data are presented as frequency (percentage). CI: confidence interval; FFRct: CT-derived Fractional Flow Reserve; ICA: invasive coronary angiography; MACE: major adverse cardiac event.				

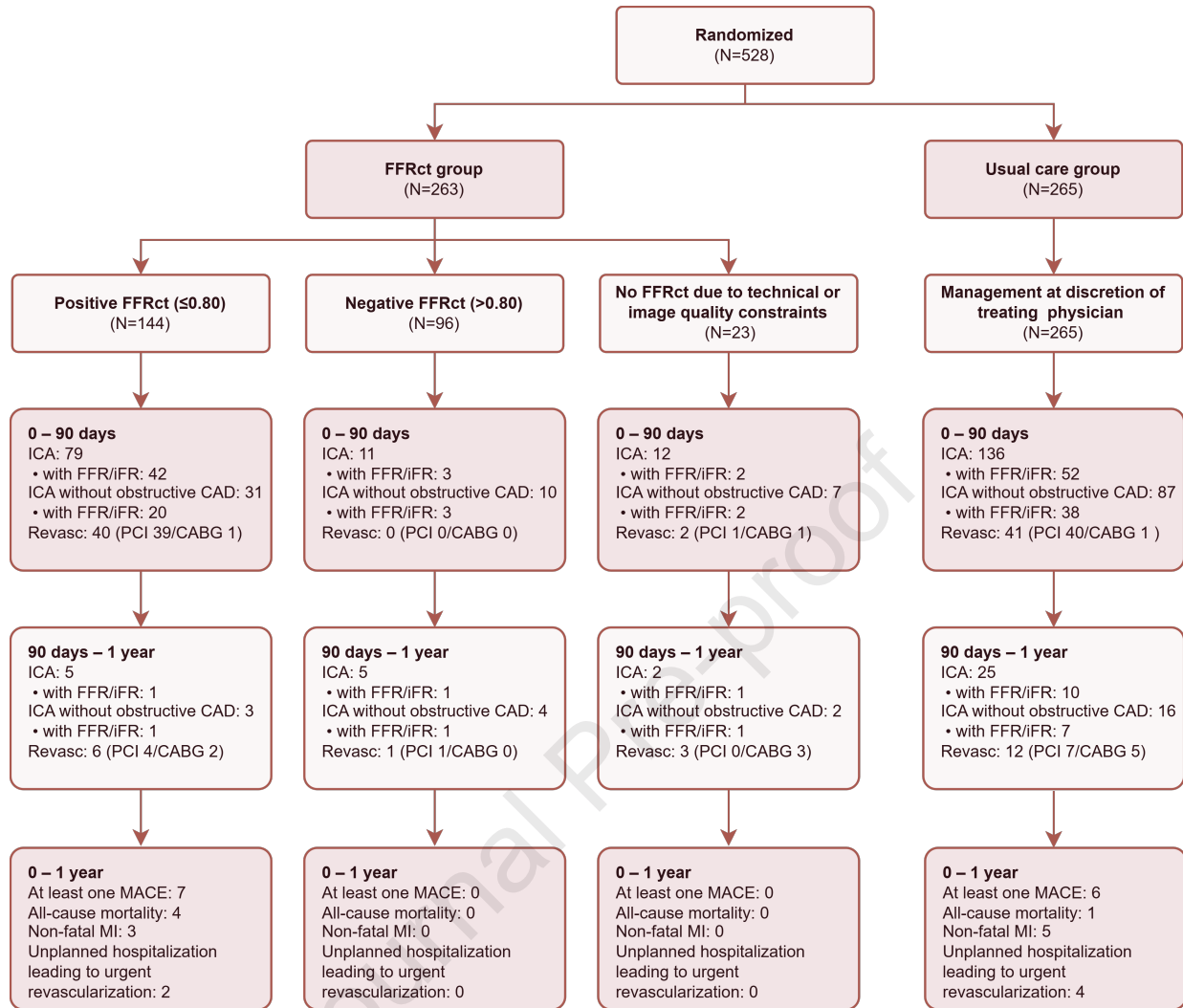
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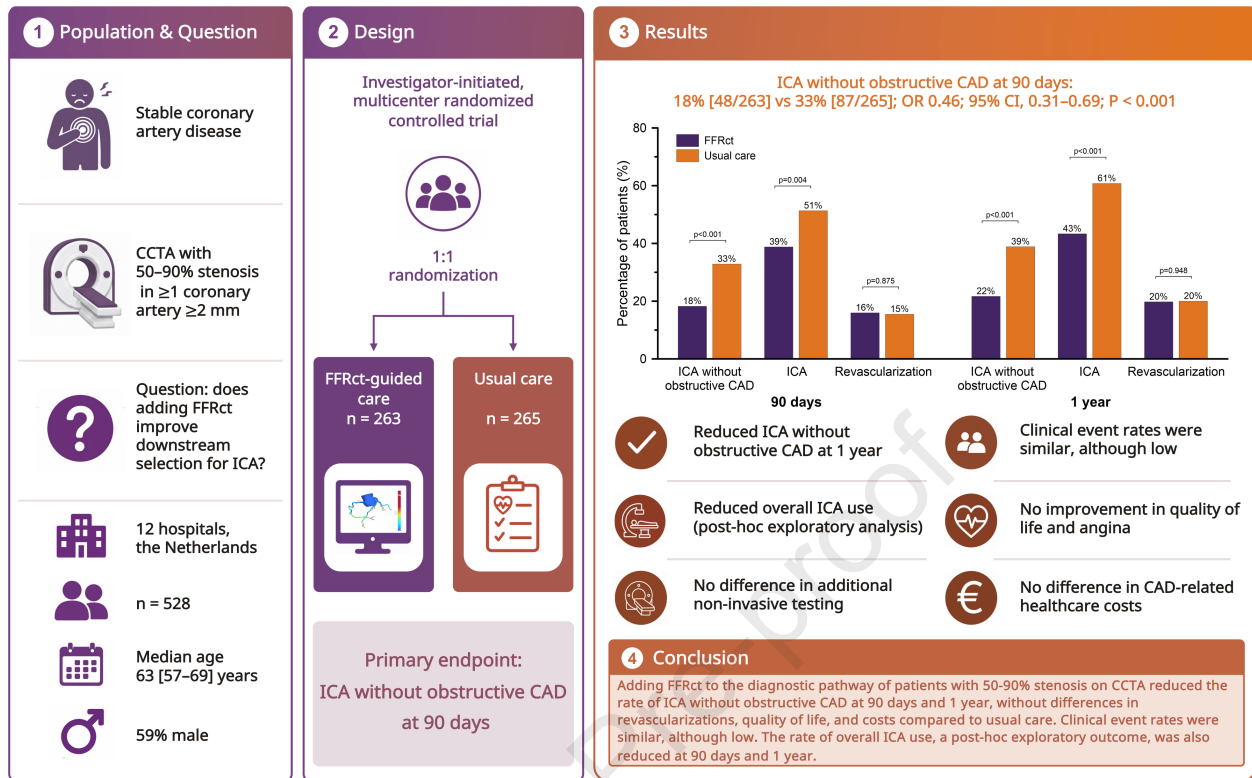


**No. at Risk**

FFRct	263	260	258	255	244
Usual care	265	263	263	261	247



Addition of FFRct in the diagnostic pathway of patients with stable CAD (FUSION Study)



Supplementary data**Trial organization**

Participating hospitals, all in the Netherlands, with the total number of patients recruited between parentheses:

Erasmus Medical Center, Rotterdam (138);
St Jansdal Hospital, Harderwijk (116);
Adrz Hospital, Goes (76);
Maastricht University Medical Center, Maastricht (50);
Albert Schweitzer Hospital, Dordrecht (39);
Martini Hospital Groningen, Groningen (28);
Haga Teaching Hospital, The Hague (20);
Slingeland Hospital, Doetinchem (19);
Hospital Gelderse Vallei, Ede (13);
Gelre Hospital, Apeldoorn (11);
University Medical Center Groningen, Groningen (9);
Franciscus Gasthuis, Rotterdam (9)

Clinical Event Committee:

Prof. F. Zijlstra, Dr. K. Mahmoud, Dr. J. F. Bulnes Muzard

Data Centre and Monitoring:

Erasmus Medical Center, Rotterdam, the Netherlands

Supplementary Table S1.

Overview of the costs of health care resource use.

Health care resource	Cost price (2025 €)	Source
<i>Diagnostic testing</i>		
Invasive coronary angiography	€ 1255	1
Invasive Fractional Flow Reserve	€ 1310	1
Invasive coronary imaging	€ 780	1, 2
Cardiovascular Magnetic Resonance	€ 296	3
PET-CT	€ 776	3
Single-photon emission CT	€ 571	3
Exercise stress ECG/echocardiography	€ 210	1, 2
Coronary CT Angiography	€ 217	3
FFRct analysis (Heartflow)	€ 1000	NA
<i>Revascularization procedures</i>		
Percutaneous coronary intervention	€ 3440	1
Coronary artery bypass graft	€ 9965	1
<i>Hospital visit/stay</i>		
Outpatient clinic visit	€ 133	3
Hospital stay (per day)	€ 714	3
Stay at intensive care unit (per day)	€ 3022	3
<i>Cardiovascular medication (costs per day, based on standard daily dosage)</i>		
ACE-I/angiotensin receptor blocker	€ 0.09	4, 5
Beta-blocker	€ 0.14	4, 5
Calcium antagonist	€ 0.16	4, 5
P2Y12 receptor inhibitors	€ 0.24	4, 5
Anticoagulation	€ 2.06	4, 5
Diuretics/aldosterone receptor antagonist	€ 0.06	4, 5
Digoxin	€ 0.06	4, 5
Long-acting nitrate	€ 0.38	4, 5
Salicylates	€ 0.11	4, 5
Statin	€ 0.09	4, 5
Cholesterol absorption inhibitor	€ 0.11	4, 5
ACE-I: angiotensin-converting enzyme inhibitor; CT: computed tomography; FFR: fractional flow reserve; FFRct: Computed Tomography-derived Fractional Flow Reserve; NA: not applicable; PET-CT: positron emission tomography-Computed Tomography;		

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Supplementary Table S2.

Per-protocol analysis and sensitivity analysis for the primary endpoint of ICA without obstructive CAD at 90 days

Endpoint	FFRct	Usual care	Unadjusted OR (95% CI)	P-value	Adjusted OR (95% CI)*	P-Value	Adjusted OR (95% CI)**	P-Value
Per-protocol analysis: excluding patients in whom FFRct result was not incorporated	N=256	N=265						
ICA without obstructive CAD at 90 days	41 (16)	87 (33)	0.39 (0.26-0.59)	<0.001	0.40 (0.26-0.61)	<0.001	0.38 (0.24-0.58)	<0.001
Sensitivity analysis: excluding patients with unavailable or not incorporated FFRct results	N=233	N=265						
ICA without obstructive CAD at 90 days	34 (15)	87 (33)	0.35 (0.22-0.55)	<0.001	0.36 (0.23-0.56)	<0.001	0.33 (0.21-0.53)	<0.001

Data are presented as number, frequency (percentage) or as odds ratio with 95% confidence interval.
 *Odds ratio was adjusted for age, sex, ln(coronary artery calcium score + 1), and type of chest pain.
 **Odds ratio was adjusted for age, sex, ln(coronary artery calcium score + 1), type of chest pain, and center.
 CI: confidence interval; FFRct: Computed Tomography-derived Fractional Flow Reserve; ICA: invasive coronary angiography; OR: odds ratio.

Supplementary Table S3.

Invasive coronary angiography rates at 90 days in the FFRct and usual care group in different subgroups based on Coronary Computed Tomography Angiography characteristics

Baseline CCTA characteristics	FFRct, n	ICA at 90 days, n (%)	Usual care, n	ICA at 90 days, n (%)	P-Value
Total	263	102 (39)	265	136 (51)	0.004
One-vessel disease	178	64 (36)	185	98 (53)	0.001
Two-vessel disease	59	26 (44)	51	25 (49)	0.604
Three-vessel disease and/or left main disease	26	12 (46)	29	13 (45)	0.921
Left main and/or proximal LAD	114	40 (35)	122	71 (58)	<0.001
CAC score <300	152	60 (39)	157	89 (57)	0.002
CAC score $\geq$ 300	111	42 (38)	108	47 (44)	0.392
CADRADS 3	185	62 (34)	167	78 (47)	0.012
CADRADS 4	78	40 (51)	98	58 (59)	0.295
Data are presented as numbers or frequencies (percentages) and are purely descriptive within subgroups. P-values represent within-subgroup comparisons and should not be interpreted as evidence of a different effect between subgroups. CAC: coronary artery calcium; CADRADS: Coronary Artery Disease-Reporting and Data System; CCTA: Coronary Computed Tomography Angiography; FFRct: CT-derived Fractional Flow Reserve; ICA: invasive coronary angiography.					

Supplementary Table S4.

Additional non-invasive diagnostic tests during follow-up

Non-invasive diagnostic tests	FFRct (n=263)	Usual care (n=265)	P-value
At 90 days			
At least one additional test	8 (3)	26 (10)	0.002
Exercise ECG	5 (2)	9 (3)	
Stress echocardiogram	0 (0)	1 (0.4)	
PET myocardial perfusion imaging	3 (1)	4 (2)	
CMR perfusion imaging	0 (0)	6 (2)	
SPECT	0 (0)	6 (2)	
CCTA	0 (0)	0 (0)	
At 1 year			
At least one additional test	28 (11)	37 (14)	0.246
Exercise ECG	18 (7)	18 (7)	
Stress echocardiogram	0 (0)	1 (0.4)	
PET myocardial perfusion imaging	4 (2)	6 (2)	
CMR perfusion imaging	1 (0.4)	8 (3)	
SPECT	5 (2)	8 (3)	
CCTA	2 (0.8)	0 (0)	
Data are presented as frequency (percentage). CCTA: coronary computed tomography angiography; CMR: cardiovascular magnetic resonance; ECG: electrocardiogram; PET: positron emission tomography; SPECT: single photon emission computed tomography			

Supplementary Table S5.

Medication use at baseline, 90 days and 1 year in the FFRct and usual care group

Medication	Baseline		90 days		1 year	
	FFRct (n=263)	Usual care (n=265)	FFRct (n=261)	Usual care (n=265)	FFRct (n=259)	Usual care (n=262)
Lipid-lowering	129 (49)	131 (49)	216 (83)	224 (85)	212 (82)	215 (82)
Antiplatelet*	71 (27)	83 (31)	163 (62)	159 (60)	150 (58)	155 (59)
Anticoagulation**	36 (14)	34 (13)	37 (14)	34 (13)	38 (15)	37 (14)
Data are presented as frequency (percentage).						
*Including salicylates and P2Y12 receptor inhibitors.						
** Including direct oral anticoagulants and vitamin K antagonist						

Supplementary Table S6.

Quality of life and angina status at baseline, 90 days, and 1 year

Measure	Timepoint	N	FFRct	N	Usual care	P-value
EQ VAS	Baseline	242	75 [61-85]	230	75 [60-81]	0.705
	90 days	221	75 [61-85]	229	75 [65-82]	0.480
	1 year	230	80 [69-85]	221	78 [66-86]	0.590
EQ-5D-5L index	Baseline	243	0.82 [0.73-0.89]	230	0.81 [0.73-0.89]	0.191
	90 days	221	0.85 [0.76-1.00]	231	0.85 [0.74-1.00]	0.815
	1 year	230	0.86 [0.78-1.00]	223	0.85 [0.77-1.00]	0.704
SAQ summary score	Baseline	223	72 [54 - 87]	199	71 [55 - 85]	0.754
	90 days	193	82 [65 - 96]	211	80 [64 - 94]	0.578
	1 year	207	88 [73 - 97]	192	85 [68 - 97]	0.407
SF-36 Physical Functioning	Baseline	242	75 [50 - 90]	226	75 [50 - 90]	0.917
	90 days	219	75 [55 - 90]	226	80 [60 - 95]	0.255
	1 year	228	80 [60 - 95]	219	80 [55 - 95]	0.368
SF-36 Role limitations due to physical health	Baseline	242	50 [0 - 100]	226	50 [0 - 100]	0.939
	90 days	219	75 [0 - 100]	226	75 [0 - 100]	0.784
	1 year	228	100 [25 - 100]	219	75 [0 - 100]	0.067
SF-36 Role limitations due to emotional problems	Baseline	242	100 [33 - 100]	226	100 [33 - 100]	0.912
	90 days	219	100 [33 - 100]	226	100 [33 - 100]	0.905
	1 year	227	100 [67 - 100]	219	100 [33 - 100]	0.299
SF-36 Energy/fatigue	Baseline	241	55 [40 - 70]	226	55 [40 - 70]	0.552
	90 days	219	60 [40 - 75]	226	60 [45 - 80]	0.550
	1 year	226	65 [45 - 76]	219	60 [45 - 80]	0.434
SF-36 Emotional well-being	Baseline	241	76 [60 - 86]	226	76 [60 - 88]	0.626
	90 days	219	80 [68 - 88]	226	80 [64 - 88]	0.569
	1 year	226	80 [68 - 88]	219	80 [68 - 88]	0.944
SF-36 Social functioning	Baseline	242	75 [63 - 100]	226	75 [63 - 100]	0.460
	90 days	219	75 [63 - 100]	226	88 [63 - 100]	0.101
	1 year	227	88 [63 - 100]	219	88 [63 - 100]	0.640
SF-36 Pain	Baseline	242	78 [55 - 90]	226	78 [58 - 90]	0.418
	90 days	219	78 [58 - 100]	226	78 [68 - 100]	0.132
	1 year	227	78 [68 - 100]	219	78 [58 - 100]	0.557
SF-36 General health	Baseline	241	55 [40 - 65]	226	50 [35 - 65]	0.521
	90 days	219	55 [40 - 70]	226	55 [40 - 70]	0.568
	1 year	226	60 [40 - 70]	219	55 [40 - 70]	0.502

Data are presented as median [25th – 75th percentiles]. P-values compare FFRct-guided care with usual care at each timepoint.
EQ VAS: EuroQol visual analogue scale; EQ-5D-5L: EuroQol 5-dimension 5-level; FFRct: Computed Tomography-derived Fractional Flow Reserve; SAQ: Seattle Angina Questionnaire; SF-36: 36-item Short Form Health Survey.

Supplementary Table S7.

Change from baseline in quality of life and angina status

Measure	Time interval	FFRct		Usual care		P-value
		N	Median change	N	Median change	
EQ VAS	Baseline to 90 days	211	0 [-7 - 10]	210	0 [-5 - 10]	0.788
	Baseline to 1 year	219	2 [-5 - 11]	198	1 [-5 - 10]	0.977
EQ-5D-5L index	Baseline to 90 days	212	0.00 [-0.04 - 0.10]	210	0.02 [-0.02 - 0.11]	0.043
	Baseline to 1 year	220	0.02 [-0.03 - 0.11]	199	0.02 [0.00 - 0.12]	0.340
SAQ summary score	Baseline to 90 days	176	5 [-2 - 17]	171	6 [-2 - 18]	0.493
	Baseline to 1 year	185	10 [-1 - 26]	159	9 [-1 - 23]	0.534
SF-36 Physical Functioning	Baseline to 90 days	210	0 [-5 - 10]	206	5 [-5 - 15]	0.100
	Baseline to 1 year	217	5 [-5 - 15]	196	5 [-5 - 15]	0.388
SF-36 Role limitations due to physical health	Baseline to 90 days	210	0 [0 - 25]	206	0 [0 - 25]	0.515
	Baseline to 1 year	217	0 [0 - 50]	196	0 [0 - 25]	0.121
SF-36 Role limitations due to emotional problems	Baseline to 90 days	210	0 [0 - 33]	206	0 [0 - 33]	0.761
	Baseline to 1 year	216	0 [0 - 33]	196	0 [0 - 33]	0.512
SF-36 Energy/fatigue	Baseline to 90 days	210	5 [-5 - 10]	206	5 [-5 - 15]	0.442
	Baseline to 1 year	214	5 [-5 - 20]	196	5 [-5 - 19]	0.322
SF-36 Emotional well-being	Baseline to 90 days	210	4 [-4 - 12]	206	4 [-4 - 12]	0.990
	Baseline to 1 year	214	4 [-4 - 12]	196	4 [-4 - 12]	0.874
SF-36 Social functioning	Baseline to 90 days	210	0 [-13 - 13]	206	0 [0 - 13]	0.066
	Baseline to 1 year	216	0 [0 - 25]	196	0 [0 - 13]	0.476
SF-36 Pain	Baseline to 90 days	210	0 [-3 - 11]	206	0 [-3 - 13]	0.312
	Baseline to 1 year	216	2.5 [0 - 23]	196	0 [-8 - 13]	0.131
SF-36 General health	Baseline to 90 days	210	0 [-5 - 10]	206	0 [-5 - 10]	0.967
	Baseline to 1 year	214	5 [-5 - 15]	196	3 [-5 - 10]	0.638

Data are presented as median [25th – 75th percentile]. Change from baseline was calculated only for patients with both baseline and follow-up data available for the respective questionnaire. P-values compare median change between FFRct-guided care and usual care.

EQ VAS: EuroQol visual analogue scale; EQ-5D-5L: EuroQol 5-dimension 5-level; FFRct: Computed Tomography-derived Fractional Flow Reserve; SAQ: Seattle Angina Questionnaire; SF-36: 36-item Short Form Health Survey.

Supplementary Table S8.

Mean (95% confidence interval) costs per patient in the FFRct and usual care groups, stratified by cost category. In this scenario, the cost price of FFRct was €1,000.

	FFRct (n = 263)	Usual care (n = 265)
Mean costs per category (95% CI)		
CT-derived Fractional Flow Reserve [#]	€ 913 (€ 871 – € 939)	€ 0
Additional non-invasive diagnostic testing	€ 42 (€ 27 – € 63)	€ 59 (€ 40 – € 83)
Diagnostic invasive coronary angiography	€ 870 (€ 708 – € 1,058)	€ 1,332 (€ 1,151 – € 1,525)
Percutaneous coronary intervention	€ 894 (€ 680 – € 1,156)	€ 876 (€ 660 – € 1,143)
Coronary artery bypass grafting	€ 476 (€ 197 – € 938)	€ 394 (€ 141 – € 817)
Outpatient clinic	€ 399*	€ 399*
Hospital stay	€ 38 (€ 14 – € 84)	€ 140 (€ 54 – € 348)
Cardiovascular medication	€ 209 (€ 182 – € 241)	€ 216 (€ 188 – € 249)
Mean total costs (95% CI)	€ 3,842 (€ 3,408 – € 4,391)	€ 3,415 (€ 3,006 – € 3,984)
Median total costs (25th-75th percentile)	€ 2,096 (€ 1,513 – € 4,967)	€ 2,456 (€ 673 – € 4,502)
The 95% CIs of the mean costs, both for the mean total costs and the categories, were estimated using a nonparametric bias-corrected and accelerated interval bootstrap approach with 10,000 repetitions. [#]costs FFRct: €1,000 per analysis *Outpatient clinic costs for all patients are equal to € 399. Therefore, confidence intervals could not be calculated using bootstrap method. CI: confidence interval; CT: Computed Tomography		

Supplementary Table S9.

Mean (95% confidence interval) costs per patient in the sensitivity analysis in the FFRct and usual care groups, stratified by cost category. In this scenario, the cost price of FFRct was €600.

	FFRct (n = 263)	Usual care (n = 265)
Mean costs per category (95% CI)		
CT-derived Fractional Flow Reserve [#]	€ 548 (€ 522 – € 564)	€ 0
Additional non-invasive diagnostic testing	€ 42 (€ 27 – € 63)	€ 59 (€ 40 – € 83)
Diagnostic invasive coronary angiography	€ 870 (€ 708 – € 1,058)	€ 1,332 (€ 1,151 – € 1,525)
Percutaneous coronary intervention	€ 894 (€ 680 – € 1,156)	€ 876 (€ 660 – € 1,143)
Coronary artery bypass grafting	€ 476 (€ 197 – € 938)	€ 394 (€ 141 – € 817)
Outpatient clinic	€ 399*	€ 399*
Hospital stay	€ 38 (€ 14 – € 84)	€ 140 (€ 54 – € 348)
Cardiovascular medication	€ 209 (€ 182 – € 241)	€ 216 (€ 188 – € 249)
Mean total costs (95% CI)	€ 3,477 (€ 3,044 – € 4,030)	€ 3,415 (€ 3,006 – € 3,984)
Median total costs (25th-75th percentile)	€ 1,696 (€ 1,114 – € 4,567)	€ 2,456 (€ 673 – € 4,502)
The 95% CIs of the mean costs, both for the mean total costs and the categories, were estimated using a nonparametric bias-corrected and accelerated interval bootstrap approach with 10,000 repetitions. [#]costs FFRct: €600 per analysis *Outpatient clinic costs for all patients are equal to € 399. Therefore, confidence intervals could not be calculated using bootstrap method. CI: confidence interval; CT: Computed Tomography		